

IoT SuperServer ARS-E103-JONX-H

Compact fanless edge AI platform, with 8-core Jetson Orin™ NX module. Up to 157 TOPS.

Key Applications

Imaging and Computer Vision, Industrial Automation, Healthcare, Artificial Intelligence, Digital Signage / PoS,

Key Features

- Support Super Mode 40W boosting up AI inferencing performance;
- Fanless DIN Rail Mounted System
Jetson Orin NX Fanless Edge AI Inferencing System;
- 8-core Arm® Cortex® -A78AE v8.2 64-bit CPU 2MB L2 + 4MB L3;
- 1024-core NVIDIA Ampere GPU with 32 Tensor Cores, Up to 157 TOPS;
- 1 10G/1G LAN optional by Sku (10G: -H(2), 1G: -L(2))
4 1Gbe w/ POE or 2 x1Gbe w/ PoE+ (by AOM-POE-001 jumper setting);
- 1 M.2 B-key 3042/3052 (USB3.2, USB2.0) w/ dual nano-SIM
1 M.2 M-key 2280 (PCIe4.0 x4)
1 M.2 E-key 2230 (PCIe x1, USB2.0) w/ optional WiFi module&antenna kit;
- 1 NVMe 2280 at M.2 M-Key;
- Lockable DC-IN Power Connector, supporting DC 9-36V
Operating Temperature -25 to 60 (-13°F to 140°F) with 0.7m/s airflow and Industrial Grade peripherals;



Form Factor	Fanless Embedded Enclosure: 185 x 80 x 140mm (7.28" x 3.15" x 5.5") Package: 290 x 200 x 230mm (11.42" x 7.87" x 9.06")
Processor	Arm® Cortex®-A78AE v8.2 64-bit CPU 8C/8T; 6MB Cache
System Memory	Onboard Memory 16GB ECC LPDDR5X
Drive Bays Configuration	1 M.2 PCIe 4.0 x4 NVMe slot (M-key 2280(default))
Expansion Slots	1 M.2 PCIe 4.0 x1 slot (E-key 2230(default); USB 2.0) 1 M.2 slot (B-key 3052(default)/3042; USB 3.0)
On-Board Devices	3 RJ45 1GbE with Intel® I210-IT 1 RJ45 10GbE with Marvell® AQC113C 1 RJ45 1GbE with SoC No NIC option supported
Input / Output	LAN: 3 RJ45 1 GbE LAN ports (Intel® I210-IT) 1 RJ45 10 GbE LAN port (Marvell® AQC113) 1 RJ45 1 GbE LAN port (SoC) USB: 1 USB 3.2 Gen2 Type-C port(Front) 3 USB 3.2 Gen2 Type-A ports(Front) Video: 1 HDMI 2.0b port(Front) Serial: 2 COM RS232/RS422/RS485 ports(Front) Others: 1 CANBus port 1 4 DI/4 x DO (terminal block) port 2 nano SIM holder ports



System Cooling	Fans: N/A
Power Supply	1x 180W power supply
Dimensions and Weight	Weight: Gross Weight: 4.75 lbs (2.15 kg) Net Weight: 3.25 lbs (1.47 kg) Available Color: Black
Operating Environment	RoHS Compliant Operating Temperature: -25°C to 60°C (-13°F to 140°F) with 0.7 m/s airflow Non-operating Temperature: -40°C to 60°C (-40°F to 140°F) Operating Relative Humidity: 8% to 80% (non-condensing) Non-operating Relative Humidity: 8% to 90% (non-condensing)
Motherboard	Super AOM-JSOR-001
Chassis	CSE-E102L2